

Polarization Analysis

The method of the data analysis and polarization calculation is discussed here.

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Data

For each helicity window the current state of the scalers is written down. The writing down is triggered by the arrival of a new helicity gate. About 1ms is assigned to this process by a gate generator and during this time the counting of the scaler is turned off. The scaler accumulates the counters during one run of data taking. Sometimes, the scaler memory bits are corrupted by radiation in the hall. In order to make the counting more reliable, two parallel scalers are used. The raw data from CODA are read in and converted to an NTUPLE format. The scalers count the following signals (the order is arbitrary):

1. Left arm, $(LG \times Ap)_L$ (coincidence of the Lead class and aperture detectors)
2. Right arm, $(LG \times Ap)_R$
3. Coincidence $(LG \times Ap)_L \times (LG \times Ap)_R$
4. Accidental coincidence $(LG \times Ap)_{L, \text{delayed}} \times (LG \times Ap)_R$
5. BCM (beam current monitor)
6. LG_L delayed
7. LG_R
8. LG_{L+R}
9. Ap_L delayed
10. Ap_R
11. 100kHz pulser

The signals 3,4 and 5 are used to calculate the beam polarization, while the others are used for various checks and calibrations.

Measuring the counting rate

Verifying the consistency of the data

It is very important to make sure that the train of helicity gates is consistent and has no hole in it. A hole may happen if, for example, the DAQ suffered a dead period because of a networking interruption. Such holes indeed do occur sometimes, but not more often than a few per minute. Several helicity windows around the area of inconsistency are rejected. The tests for consistency include:

- The time between the consecutive gates is correct within several ms.
- The scaler increment for one step should have a reasonable value

Several helicity cycles around the inconsistent cycle found, are rejected.

Helicity determination

The counting rates are accumulated helicity wise. The helicity for a given cycle is taken:

- *non-delayed mode*: from the previous cycle (since the states of the scalers along with the helicity flag are read out when the next cycle just arrives);
- *delayed mode*: 8 cycles later (in this case the data on the cycles are stored in a 8-word buffer, till the helicity state is decided)

Asymmetry calculation

For each cycle, the increments of the scalers with respect to the previous cycle are calculated, and accumulated for the given helicity. In order to calculate the asymmetry the counting rates of 2-arm coincidence $N(\text{coin})_+$, $N(\text{coin})_-$, is used:

- normalization to the number of cycles: $N(\text{coin}_1)_+ = N(\text{coin})_+ / N(\text{cycles})_+$
- subtraction of the accidental coincidence rate: $N(\text{coin}_2)_+ = N(\text{coin}_1)_+ - N(\text{accid}_1)_+$
- normalization to the beam flux: $N(\text{coin}_3)_+ = N(\text{coin}_2)_+ / N(\text{BCM}_1)_+$
- calculate the asymmetry: $A = (N(\text{coin}_3)_+ - N(\text{coin}_3)_-) / (N(\text{coin}_3)_+ + N(\text{coin}_3)_-)$